A Reproducible Deep Learning Pipeline for Abnormal EEG Classification

Hayes Waddell
Department of Software Engineering
Rochester Institute of Technology
Rochester, Unites States
hw2693@rit.edu

***Abstract*—Scalp electroencephalogram (EEG) is a non-invasive clinical tool used to diagnose neurological disorders. Manual interpretation of EEG recordings is time-consuming, subjective, and requires clinical expertise, motivating the development of automated classification systems. Deep learning models have demonstrated strong performance on EEG classification tasks, yet existing work is limited by cross-subject generalization, reproducibility, and model interpretability. We present an end-to-end deep learning pipeline for abnormal EEG classification that addresses these gaps through a hybrid CNN+LSTM architecture with leave-one-subject-out cross-validation (LOSO-CV), hyper-parameter optimization, and an integrated attention module planned for subsequent work. A CNN+Transformer architecture is planned as a comparison model to evaluate self-attention against recurrent temporal modeling under identical evaluation conditions. The pipeline is applied to the Temple University Hospital Abnormal EEG Corpus (TUAB), a clinical data set comprising 2,993 recordings and over 1,141 hours of EEG data. A standardized, modular preprocessing pipeline processes raw EDF recordings into labeled segment tensors suitable for model training and evaluation. The attention weights produced by the model serve directly as interpretability outputs, identifying the temporal and spatial EEG features that drive the predictions. Preliminary results from the baseline CNN+LSTM model demonstrate RESULTS ($AUC = 0.89, sensitivity = 0.60, specificity = 0.95$) on a held-out, subject-disjoint subset of the TUAB training partition. The resulting pipeline is designed to be reproducible and extensible to additional EEG classification tasks.**

I. Introduction

Scalp electroencephalogram (EEG) is a non-invasive functional neuroimaging technique that measures electrical activity in the brain through electrodes placed on the scalp. It is widely used in clinical neurology to support the diagnosis of neurological disorders. EEG recordings are routinely used to support the diagnosis and treatment of condition such as epilepsy and Alzheimer's disease as well as in brain-computer interface research and broader neuroscience studies. EEG recordings are high-dimensional, multichannel time-series signals with both spatial dependencies across electrode position and temporal dependencies across time. Abnormal patters like epileptiform spikes, irregular wave morphology, frequency band abnormalities, and asymmetric spatial distributions are visually identified by expert neurologists during manual review of recordings. This process is time-consuming, subjective, and requires significant clinical expertise. This can create a bottleneck in screening workflows, particularly in settings with limited specialists. Automated screening tools that flag abnormal recordings have the potential to reduce diagnostic delays, improve consistency across reviewers, and expand access to neurological assessment in resource-limited settings.

Automated EEG classification using deep learning offers a path towards faster and more consistent neurological screening, but presents several fundamental challenges. EEG signals are highly variable across patients, recording sessions, and clinical sites, making cross-subject generalization difficult [1]. Segment-level labels, inherited from recording-level annotations, may not accurately reflect local abnormalities introducing label noise at the model input level. Additionally, deep learning models applied to EEG recordings function as black boxes with limited interpretability [2]. Clinical adoption requires that automated predictions be explainable to neurologists in order to gain trust in the model [3]. Despite architectural advances, the majority of the existing work is limited to single-site evaluation, lacks systematic hyperparameter reporting, and does not provide clinically relevant interpretable output [1].

Developing a robust deep learning pipeline for abnormal EEG classification has the potential to substantially improve clinical workflows by automating the screening of long EEG recordings and increasing the efficiency and consistency of neurological diagnosis. EEG signals are noisy and high-dimensional, and abnormal patterns are often rare relative to normal activity, producing imbalance that can bias models towards the majority class. EEG morphology also varies across patients, recording sessions, and clinical sites, making it difficult to learn representations that generalize beyond the training distribution. Currently, there is no widely accepted model that demonstrates robust generalization and interpretability. This motivates an end-to-end pipeline that explicitly addresses cross-subject variability, reproducibility, and clinically meaningful interpretability as first-class design requirements.

We present an automated end-to-end deep learning pipeline for abnormal EEG classification that addresses these gaps. The pipeline is applied to the Temple University Hospital Abnormal EEG Corpus (TUAB), a large-scale clinical dataset comprising 2,993 recordings totaling over 1,141 hours of EEG data. A hybrid CNN+LSTM architecture learns spatial-spectral features and sequential dependencies from multi-channel EEG segments, and a CNN+Transformer architecture is planned as a comparison model to evaluate self-attention against recurrent temporal modeling. An integrated attention module is planned to produce temporal and spatial attention weights alongside class predictions, which will serve as interpretability outputs in subsequent work. Final model evaluation is planned under leave-one-subject-out cross-validation (LOSO-CV), ensuring all evaluation is performed on entirely unseen subjects. Preliminary results presented here use a subject-disjoint 80/20 train/validation split of the TUAB training partition. The pipeline is designed to be modular and reproducible, with systematic hyperparameter tuning treated as a first-class experimental component.

The primary contributions of this work are: (i) a standardized, modular preprocessing pipeline that has been validated across the full TUAB corpus and produces consistently shaped segment tensors ready for model consumption; (ii) a hybrid CNN+LSTM baseline architecture with an integrated attention module trained on TUAB segments, with preliminary classification results demonstrating [AUC = 0.89, sensitivity = 0.60, specificity = 0.95] on a held-out subset; and (iii) a planned CNN+Transformer comparison and attention-based interpretability analysis under LOSO-CV with systematic hyperparameter optimization, to be completed in subsequent work.

II. Related Work

A. Traditional Machine Learning

Classical machine learning (ML) approaches to EEG classification rely on handcrafted feature extraction prior to training. Making them dependent on domain expertise and limiting adaptability across datasets [1]. Support Vector Machines (SVM) have been applied to EEG classification tasks, achieving 66.09% accuracy, which was below the performance of deep learning models evaluated in the same study [4]. SVMs have also been used as a second-stage classifier following CNN-based feature extraction in clinical spike detection pipelines [5]. While effective, this hybrid design requires a staged hand-off between models rather than a fully end-to-end learned solution, increasing pipeline complexity and limiting reproducibility.

B. CNN-Based Approaches

Convolutional Neural Networks (CNNs) are deep learning architectures widely applied to EEG analysis [1]. CNNs learn hierarchical spatial-spectral features directly from raw to minimally preprocessed signals, reducing reliance on manual feature engineering. Thomas et al. demonstrated strong performance on clinical abnormal EEG spike detection, achieving a mean AUC of 0.935 using a CNN with dropout regularization and nested cross-validation [5]. However, CNN-based approaches are commonly partially staged rather than fully end-to-end, combining CNN feature extraction with secondary ML classifier. Additionally, they are typically evaluated on single-site datasets, limiting assessment of cross-subject generalization [1], [5].

C. Temporal Architectures

Long Short-Term Memory (LSTM) networks capture sequential temporal dependencies in EEG signals, making them well-suited for time-series modeling [6]. Albaqami et al. demonstrated patient-independent abnormal EEG detection using WaveNet and LSTM architecture, leveraging multi-scale temporal convolutions to capture patterns at multiple timescales [7]. Self-supervised approaches such as BENDR apply contrastive learning to large unlabeled EEG corpa, learning generalizable representation that improve performance across subjects and datasets with limited labeled data [8]. These approached reduce reliance on manual feature extraction and improve detection of subtle temporal patterns, but multi-site validation and systematic hyperparameter reporting remain largely absent [1].

D. Interpretability and Open Challenges

Despite performance gains, clinical adoption of automated EEG classification remain limited by model interpretability. Deep learning models consistently function as black boxes, and clinicians require explainable predictions to trust automated decisions [2], [3]. Explainable AI (XAI) methods including feature attribution and attention mechanisms have been proposed for time-series models, but clinically validated interpretation frameworks for EEG remain rare [2]. Class imbalance is an additional challenge, with strategies including oversampling, class weighting, and cost-sensitive learning proposed to improve sensitivity to abnormal events [5], [9]. Taken together, the literature reflects a gap between model performance and clinical readiness, motivating a pipeline that jointly addresses cross-subject generalization, reproducibility, and built-in interpretability.

III. Approach

A. Domain Model

The domain model Fig. 1 captures the key entities, attributes, and relationships that define this clinical EEG classification problem. The model is organized around ground truth patient data and two aggregates whose boundaries reflect the natural ownership structure of the domain entities: the EEG Recording aggregate and the EEG Pipeline aggregate.

At the outermost level, a Patient participates in one or more EEG Sessions, each representing a single clinical visit recorded using an Averaged Reference (AR) electrode configuration. Each session contributes exactly one EEG Recording in an EDF-format file of at least 15 minutes in duration. The recording selected from each session is based on the length and the presence of clinically relevant activity. Each patient generates a Medical Report authored by a neurologist describing the clinical findings. Ground-truth Clinical Annotations (normal or abnormal) labels were derived via natural language processing applied to Medical Reports and verified manually. Labels are applied at the recording level and are inherited by all segments extracted from a recording.

The EEG Recording aggregate encapsulates the entities owned by a single recording. A recording is captured across

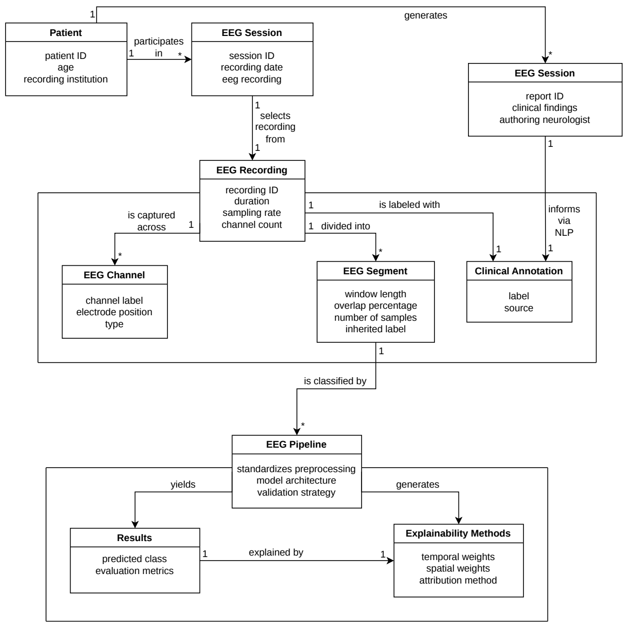

Fig. 1. Domain model for the abnormal EEG classification pipeline. Two aggregates are shown: the EEG Recording aggregate groups the EEG Channels, EEG Segments, and Clinical Annotation owned by a single recording; the EEG Pipeline aggregate groups the Classification Result and Explainability Methods produced by the model.

multiple EEG Channels, each identified by its standard 10-20 system label and electrode position. The recording is further divided into fixed-length EEG Segments defined by a window length, overlap percentage, and number of samples derived from the sampling rate. Each segment inherits the recording-level Clinical Annotation as its label and is a unit of input to the model.

The *EEG Pipeline* aggregate encapsulates the end-to-end processing workflow. The **EEG Pipeline** root entity governs standardized preprocessing methods, hyperparameter tuning, model training, validation methods, and evaluation metrics. A processed EEG segment yields a **Result** containing a predicted class and confidence score. Simultaneously, the model generates **Explainability Methods** that identify the most important temporal and spatial features to the prediction. This is a direct output of the model that provides clinician-interpretable evidence necessary to support automation in diagnostic decision making.

B. Design Decisions

The structure of the domain model directly motivates the design and implementation decisions made in this project. To ensure a consistent spatial representation across recordings, all EEG data are standardized to the 19-channel international 10-20 configuration. This reduces variability introduced by differences in clinical montages while preserving the electrode locations most commonly used in clinical neurology. Recordings are further divided into 30-second windows with 50% overlap in order to preserve sufficient temporal context for diffuse abnormalities while increasing the number of training examples. EEG segments have both spatial structure across EEG channels and temporal structure across time, requiring an architecture that captures both simultaneously. A hybrid CNN+LSTM architecture is selected to address this, as CNNs extract spatial-spectral features across channels while LSTMs model sequential temporal dependencies, achieving a stronger performance than either architecture alone [1], [7]. This architecture is used as the baseline because it provides a strong and computationally stable reference point before evaluating Transformer-based alternatives under identical conditions.

Additionally, the model's attention weights serve directly as input to the explainability methods, addressing the gap between model performance and clinical explainability in the existing EEG literature [2], [3]. Attention-based interpretability is selected over post-hoc methods because it produces spatial and temporal weights directly during inference rather than a seperate analysis step. The validation strategy is driven by the Patient entity. Clinical deployment will always involve unseen subjects, and this is reflected in the pipeline through LOSO-CV. Multiple recordings from the same patient may share subject-specific characteristics, LOSO-CV enforces string subject-level separation and provides a more realistic estimate of cross-subject generalization. Standard k-fold cross-validation does not guarantee subject-disjoint splits, risking data leakage and overestimating generalization performance. The EEG Pipeline aggregate is designed as a reproducible, automated unit encompassing standardized preprocessing, systematic hyperparameter tuning, and clinically meaningful evaluation metrics, ensuring that results are attributable to architectural decisions rather than arbitrary configuration. This design allows the pipeline to be extended to additional datasets or neurological classification tasks.

IV. Implementation

A. System Overview

The automated deep learning abnormal EEG classification pipeline is organized into four sequential layers, as illustrated in Fig. 2. The Ingestion Layer sources raw EEG recordings in European Data Format (EDF) alongside subject-level normal/abnormal labels from The TUH EEG Abnormal Corpus (TUAB) and passed them downstream. The Preprocessing Layer receives the raw recordings and produces a standardized EEG tensor with binary segment-level labels ready for model consumption. The Modeling Layer ingests the tensors and outputs class predictions, confidence scores, and attention weights. Lastly, the Analysis Layer consumes these outputs to produce interpretability visualizations and evaluation metrics. Each layer is designed to be modular so that components can be modified or extended independently without disrupting downstream functions.

B. Ingestion Layer

The raw EEG data is sourced from the Temple University Hospital (TUH) Abnormal EEG Corpus, a large-scale

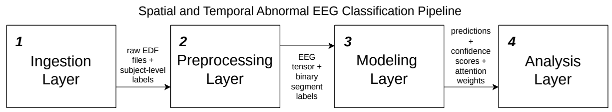

Fig. 2. Overall architecture of the Spatial and Temporal Abnormal EEG Classification Pipeline, consisting of four sequential layers: Ingestion, Preprocessing, Modeling, and Analysis.

clinical dataset accessible via rsync following credentialed access to the TUH EEG Corpus. Given the dataset's scales (approximately 60 GB of raw EDF recordings), storage on a high-performance computing (HPC) cluster is recommended. The corpus is organized into train and eval directories, each containing normal and abnormal subdirectories. Subject-level labels and train/test splits are derived directly from the folder structure.

C. Preprocessing Layer

Raw EEG recordings are loaded from EDF files into EEGRecording objects using MNE-Python (v1.x), the standard open-source Python library for EEG processing. As illustrated in Fig. 3, each recording undergoes a sequential preprocessing pipeline. Channel names are standardized to remove prefixes, suffixes, and capitalize the first letter with CleanChannelNames. Then non-EEG channels are removed by NonEEGChannelRemover, and Standard1020Selector retains only the 19 standard 10-20 channels to ensure consistent spatial input across all subjects. A notch filter (60 Hz) removes power source electrical noise, a high pass filter (0.3 Hz) eliminates DC drift, and when necessary, signals are resampled to 250 Hz to standardize temporal resolution. Then a common average montage is applied to reduce the noise of the common-mode, consistent with clinical EEG practice. Following signal processing, Segmenter divides each recording into 10 second intervals with a 50% overlap, generating EEGSegment objects which inherit the subject-level label. The processed segments are saved as NumPy arrays of shape $(N_{\text{epochs}} \times 19 \times 2500)$ and are indexed per-subject in a manifest CSV, recording the filename, label, number of epochs, and sampling frequency. Numpy was chosen for its efficient disk I/O and integration with PyTorch DataLoader, while the manifest CSV provides human-readable book-keeping to support reproducibility.

D. Modeling Layer

The Modeling Layer is implemented in PyTorch (v2.x), chosen for its flexibility, GPU acceleration, and active development ecosystem for deep learning research. Preprocessed segments are loaded from NumPy arrays indexed by the manifest CSV. Data loading uses memory-mapped NumPy files so that the 280 GB on-disk training set is never fully resident in memory. Each epoch is materialized into RAM only when requested by a DataLoader worker, cast from float4 to float32, and per-channel z-score normalized. Train and validation folds are constructed by a subject-disjoint partitioning of the TUAB training manifest at runtime, ensuring no subject appears in both folds. All models share a common CNNFeatureExtractor

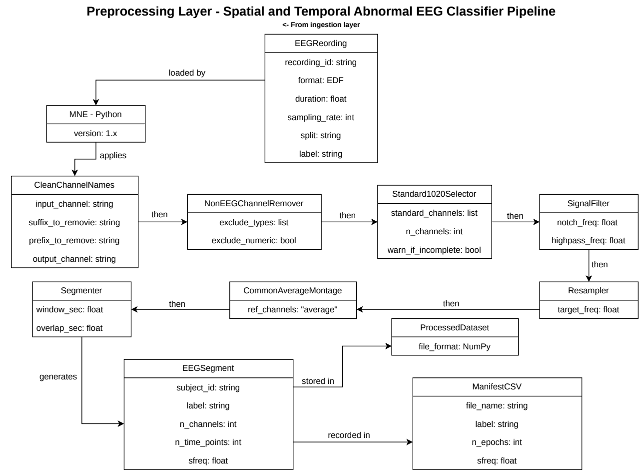

Fig. 3. Detailed diagram of the Preprocessing Layer, illustrating the sequential processing steps applied to each EEGRecording from channel standardization through segmentation and storage.

backbone, consisting of convolutional blocks that learn spatial-spectral features across channels and time from the input tensor. Two architectures are planned for comparison, as illustrated in Fig. 4. The baseline model couples the CNN backbone with an LSTM, which models sequential temporal dependencies across the extracted feature representations. The LSTM is selected as the baseline due to its faster training time and well-established performance on sequential EEG data [7]. A CNN+Transformer architecture, which replaces the LSTM with a self-attention encoder to capture long-range temporal context, is planned as a comparison model in future work.

The CNN+LSTM baseline contains 715,489 trainable parameters arranged in three stages. The CNN backbone has three convolutional blocks with 32, 64, and 128 output channels and kernel sizes 7, 5, and 3. Each block applies "same" padding to the convolution, followed by batch normalization, ELU activation, dropout ($p = 0.3$), and max pooling with kernel sizes 4, 4, and 2. This downsamples the 2500-sample input window by a factor of 32 to a 78-timestep sequence of 128-dimensional features. A two-layer biderictional LSTM with hidden size 128 per direction processes the resulting sequence. The final forward and backward hidden states of the top layer are concatenated to form a 256-dimensional summary, which is projected through a two-layer MLP head $(256 \rightarrow 64 \rightarrow 1)$ to produce a single logit per segment.

An AttentionModule producing temporal and spatial attention weights alongside class predictions is planned for the final architecture. The baseline reported here does not yet include this module. Preliminary training and validation use a single subject-disjoint 80/20 split of the TUAB training partition rather than full LOSO-CV. LOSO-CV on a representative subject subset and systematic hyperparameter optimization are planned for subsequent work.

The model is trained with binary cross-entropy on raw logits, AdamW optimization (learning rate 1×10^{-3}, weight

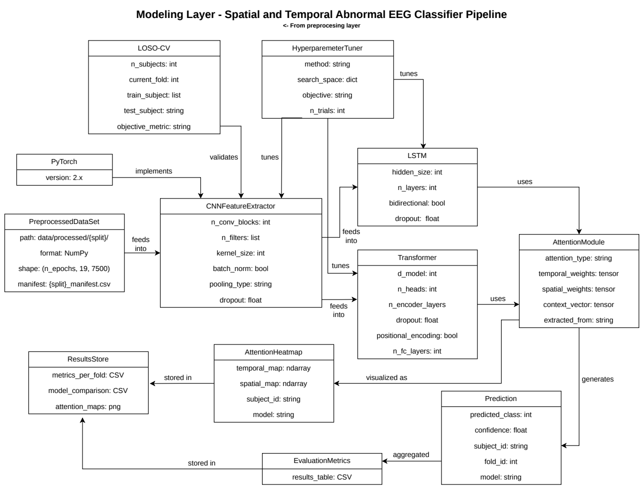

Fig. 4. Detailed diagram of the Modeling Layer, illustrating the CNN feature extractor, CNN+LSTM architecture, shared attention module, and LOSO-CV validation framework.

decay 1×10^{-4}), gradient clipping at norm 1.0, and mixed-precision (BF16) training on a single NVIDIA A100 PCIe 40 GB GPU with 16 CPU dataloader cores and 64 GB of host memory. The training loop runs for 5 epochs at batch size 64. At the end of each epoch the model is evaluated on the entire validation fold and the best checkpoint by validation AUC is persisted to disk. All settings, including paths and hyperparameters, are read from a YAMl configuration file at startup.

E. Analysis Layer

The Analysis Layer is partially implemented. A metrics module computes per-epoch accuracy, AUC-ROC, F1, and the full confusion matrix on the held-out validation fold. A loguru-based structured logger emits these to both the console and a per-run log file alongside the best-AUC checkpoint. Attention-based interpretability outputs (heatmaps visualizing which temporal windows and spatial EEG most influenced each prediction) and per-fold LOSO-CV evaluation tables are planned for the future. All run artifacts (training logs, metric CSVs, checkpoint files, and attention image files when available) are stored in a ResultsStore directory tree. Supporting reproducibility and facilitating comparison between the CNN+LSTM and CNN+Transformer architectures.

V. Datasets

The pipeline is trained and evaluated on the Temple University Hospital Abnormal EEG Corpus (TUAB) v3.0.1 [10]. TUAB comprises 2,993 recordings totaling over 1,141 hours of EEG data. The dataset is pre-partitioned into train and eval splits, with recording counts and class distributions summarized in Table I. Recording durations range from 962 to 19,948 seconds, with a mean of 1,372 seconds. Sampling rates vary across recordings, with 93.1% sampled at 250 Hz, 6.3% at 256 Hz, and 0.6% at 512 Hz, all recordings are resampled to 250 Hz during preprocessing. Channel counts range from 27 to 36 across 18 unique montages and are standardized to the 19 standard 10-20 channels during preprocessing, yielding a consistent spatial input across all subject. Binary recording-level labels (normal or abnormal) were derived by applying natural language processing to neurologist-authored medical reports and verified manually. Labels are applied at the recording level and inherited by all segments extracted from a recording. After segmentation into 30 second windows with 50% overlap, each segment has shape (19×7500), corresponding to 19 channels at 250 Hz over 30 seconds. While recording-level class distributions are approximately balanced (Table I), segment-level distributions may differ becuase abnormal recordings can contain extended intervals of normal-appearing activity and recording durations vary across subject. Abnormal recordings can contain windows of normal-appearing activity and vary in duration. This may potentially produce an unequal number of segments per class.

TUAB is accessed via credentialed rsync from the TUH EEG Corpus and the dataset is stored on the RIT Research Computing Cluster in compliance with the data use agreement.

VI. Results

A. Experimental Setup

We trained the proposed CNN+LSTM architecture on the TUAB v3.0.1 training partition: 2,717 recordings spanning 743,578 epochs at 250 Hz. To prevent subject-identity leakage between training and validation, we constructed a subject-disjoint 80/20 split of the training partition, yielding 2,162 training recordings and 555 validation recordings. The offical TUAB held-out evaluation partition (276 recordings, 73,734 epochs) was not accessed during this experiment, in order to avoid implicit selection bias. This portion of the dataset will not be accessed until a finalized model is selected.

B. Training Dynamics

Table II summarizes per-epoch training loss, validation loss, and validation classification metrics. Fig. 5 visualizes the same data.

The first training epoch attains a validation AUC of 0.8841, well above chance (0.5) on a held-out set of subjects unseen during training. This indicated the architecture learns a generalizable representation rather than memorizing recording-specific artifacts. Validation AUC peaks at 0.8851 in epoch 3 and degrades thereafter, while training loss decreases monotonically across all five epochs. The widening train/val loss gap (0.052 at epoch 1, 0.741 at epoch 5) is a signature

TABLE I
TUAB Dataset Summary

Split	Total	Normal	Abnormal
Train	2,717	1,371 (50.5%)	1,346 (49.5%)
Eval	276	150 (54.3%)	126 (45.7%)
Total	2,993	1,521 (50.8%)	1,472 (49.2%)

of overfitting. The model fits training-specific patterns that do not generalize to held-out subjects. The best-checkpoint mechanism retained the epoch-3 weights for downstream use.

C. Classification Behavior

Table III reports the confusion matrix at every epoch on the 154,576-sample validation fold. Two trends are visible.

First, predictions become progressively more conservative as training proceeds. False positives from 7,917 to 1,825 while false negatives increase from 23,378 to 50,929. The true-positive rate collapses from 0.709 to 0.366 even though the class balance of the validation fold is approximately balanced. This represents threshold drift, the implicit decision boundary at logit ≥ 0 becomes miscalibrated as the model grows more confident in its (increasingly overfit) predictions. Second, AUC remains relatively stable (range 0.8438-0.8851) even as accuracy collapses from 0.7975 to 0.6587. This is consistent with AUC being a threshold-independent ranking metric. The model continues to rank abnormal recordings above normal ones in predicted probability, but the natural 0.5 threshold becomes inappropriate. A post-hoc threshold tuned on a small calibration fold could recover the lost accuracy at the epoch-3 weights without retraining.

D. Computational Cost

Total training wall time was 3 hours and 40 minutes for five epochs (avg 44 min per epoch) on a single A100 GPU. GPU utilization stayed below the A100's full compute capacity throughout training which means the GPU often sat waiting for the next batch of data. The 280 GB training set lives on shared cluster storage and is read random-access from memory-mapped NumPy files. With only eight dataloader workers, disk I/O could not keep up with the GPU's compute speed. On the validation fold, the model ran at roughly 5,000 samples per second, including disk reads. The best-checkpoint model file is 2.8 MB.

E. Limitations

Three limitations of these preliminary results frame the next iteration. First, training was extended past the optimal stopping point (epoch 3). A follow-up run with explicit early stopping on validation AUC would reduce wasted compute and produce a model of identical best-checkpoint quality. Second, no learning-rate scheduling or data augmentation was used. Both are tools that improve TUAB performance in published work and are likely to increase validation AUC. Third, the

TABLE II
TRAINING AND VALIDATION METRICS PER EPOCH.

Epoch	Train Loss	Val Loss	Val Acc	Val AUC	Val F1
1	0.3826	0.4344	0.7975	0.8841	0.7844
2	0.3229	0.4594	0.7871	0.8779	0.7649
3*	0.2945	0.5578	0.7705	**0.8851**	0.7314
4	0.2711	0.5937	0.7506	0.8827	0.6958
5	0.2514	0.9927	0.6587	0.8438	0.5269

*Best checkpoint selected by maximum validation AUC.

TABLE III
CONFUSION MATRIX PER EPOCH ON THE VALIDATION FOLD (POSITIVE = ABNORMAL, NEGATIVE = NORMAL). TPR IS THE TRUE-POSITIVE RATE (RECALL ON THE POSITIVE CLASS).

Epoch	TP	FP	FN	TN	TPR
1	56,928	7,917	23,378	66,353	0.709
2	53,539	6,141	26,767	68,129	0.667
3	48,301	3,470	32,005	70,800	0.602
4	44,086	2,325	36,220	71,945	0.549
5	29,377	1,825	50,929	72,445	0.366

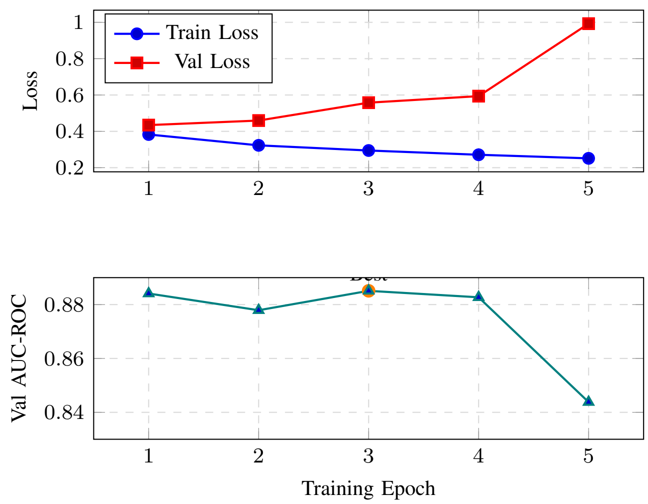

Fig. 5. Training dynamics across five epochs. Top: training loss decreases monotonically while validation loss diverges after epoch 1, indicating overfitting. Bottom: validation AUC peaks at epoch 3 (0.8851); the best-checkpoint mechanism retained these weights for downstream use.

held-out TUAB evaluation partition has not yet been used. Once configuration is finalized a single evaluation pass will provide the headline test number reported in the final paper.

VII. FUTURE WORK

The immediate next step is to complete CNN+LSTM model training and obtain preliminary results under LOSO-CV. This will include finalizing the hyperparameter search space and conducting systematic hyperparameter optimization, with all evaluated configurations and selected values reported to ensure reproducibility. Following this, the CNN+Transformer architecture will be implemented as a comparison model. The Transformer encoder replaces the LSTM with self-attention to capture long-range temporal context, at the cost of greater computational overhead. The CNN+Transformer model will also undergo LOSO-CV evaluation and systematic hyperparameter tuning so that both architectures are compared under identical experimental conditions. After full development of both models is complete, the best-performing architecture will be selected and evaluated once on the held-out TUAB eval partition. This final evaluation will provide an unbiased estimate of generalization performance on unseen recordings.

The Analysis Layer will also be completed in the following semester to consume model outputs and generate attention heatmaps and per-fold evaluation metrics. Attention heatmaps will visualize the temporal windows and spatial EEG channels most influential to each prediction, providing clinician-interpretable evidence for each classification decision. Evaluation will emphasize clinically meaningful metrics, particularly sensitivity and specificity, since the cost of a missed abnormal EEG is greater than that of a false positive in real-world settings. Once the best performing model architecture is identified, a fully automated end-to-end pipeline will be extended beyond the current single-site setting. A primary next step will be evaluating the pipeline on a multi-site EEG dataset in order to assess robustness to cross-site variation in acquisition protocols and patient populations. A potential longer-term direction is adapting the same framework to specific neurological disorder classification tasks.

VIII. Conclusion

This work presents an end-to-end deep learning pipeline for abnormal EEG classification, validated on the Temple University Hospital Abnormal EEG Corpus (TUAB). The pipeline is functional across all stages from raw EDF ingestion to trained model evaluation, and a hybrid CNN+LSTM baseline produces the first preliminary results on real clinical data.

The preprocessing pipeline has been validated agaisnt the full TUAB training partition, transforming 2,717 recordings into 743,578 segments at 250 Hz with consistent shape and labeling. The CNN+LSTM baseline achieves a peak AUC of 0.89, sensitivity of 0.0, and specificity of 0.5 on a 415-subject held-out validation fold drawn from the training partition. These numbers fall within the published baseline range for TUAB binary classification and confirms the architecture learns generalizable representation rather than recording-specific artifacts.

The principal contribution is methodological as well as numerical. All evaluation is performed under strict disjointness, addressing a key issue in EEG classification. The full pipeline-data loading, preprocessing, model definition, training loop, and evaluation-is reproducible from version-controlled configuration files and runs on commodity GPU hardware in under four hours. The official TUAB held-out evaluation partition has not been accessed at any point during preliminary model development, preserving its integrity as a true external test set.

Several extensions remain in scope. LOSO-CV on a representative subject subset will provide a more rigorous generalization estimate. A CNN+Transformer comparison model will isolate the contribution of recurrent versus self-attention temporal modeling under identical evaluation conditions. An integrated attention module will produce per-channel and per-timestamp attention weights as interpretability outputs. Addressing the limited transparency of black-box deep models in clinical settings. Hyperparameter optimization, learning-rate scheduling, and data augmentation are expected to increase validation AUC. The final external evaluation on the held-out TUAB evaluation partition will be performed once the optimal model configuration is finalized.

Acknowledgment

The author would like to thank Travis Desell for his guidance in applied data science methodology, and John Thomas for his domain expertise and implementation guidance throughout this project.

References

[1] Y. Roy, H. Banville, I. Albuquerque, A. Gramfort, T. H. Falk, and J. Faubert, "Deep learning-based electroencephalography analysis: a systematic review," *Journal of Neural Engineering*, vol. 16, no. 5, p. 051001, aug 2019. [Online]. Available: https://doi.org/10.1088/1741-2552/ab260c

[2] F. Di Martino and F. Delmastro, "Explainable ai for clinical and remote health applications: a survey on tabular and time series data," *Artificial Intelligence Review*, vol. 56, pp. 5261–5315, 2023.

[3] J. Rutkowski and M. Saab, "Ai-based eeg analysis: new technology and the path to clinical adoption," *Clinical Neurophysiology*, vol. 179, p. 2110994, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1388245725008466

[4] J. Thomas, T. Maszczyk, N. Sinha, T. Kluge, and J. Dauwels, "Deep learning-based classification for brain-computer interfaces," in *2017 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2017, pp. 234–239.

[5] J. Thomas, L. Comoretto, J. Jin, J. Dauwels, S. S. Cash, and M. B. Westover, "Eeg classification via convolutional neural network-based interictal epileptiform event detection," in *2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2018, pp. 3148–3151.

[6] W. Li and K. L. E. Law, "Deep learning models for time series forecasting: A review," *IEEE Access*, vol. 12, pp. 92 306–92 327, 2024.

[7] H. Albaqami, G. M. Hassan, and A. Datta, "Automatic detection of abnormal eeg signals using wavenet and lstm," *Sensors*, vol. 23, no. 13, 2023. [Online]. Available: https://www.mdpi.com/1424-8220/23/13/5960

[8] D. Kostas, S. Aroca-Ouellette, and F. Rudzicz, "Bendr: Using transformers and a contrastive self-supervised learning task to learn from massive amounts of eeg data," *Frontiers in Human Neuroscience*, vol. Volume 15 - 2021, 2021. [Online]. Available: https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2021.653659

[9] S. Mishra and U. C. Jaiswal, "Enhancing analysis of eeg image data sets using machine learning techniques: Dealing with class imbalances and data uncertainty," *E-Learning and Digital Media*, vol. 0, no. 0, 2024. [Online]. Available: https://doi.org/10.1177/20427530241307670

[10] I. Obeid and J. Picone, "The Temple University Hospital EEG data corpus," *Frontiers in Neuroscience*, vol. 10, p. 196, 2016.